Exercise 7: Critical Path

Critical Path:

T1 → T3 → T4 → T5 → T6 → T7

List of Activities:

T1	**Compliance framework**
T2	**Database design**
T3	**Basic functionality**
T4	**Implement modules**
T5	**UI design**
T6	**Client consultation**
T7	**Testing**

Expected Project Duration:

T1 → T3 → T4 → T5 → T6 → T7 =

7 + 10 + 11 + 8 + 2 + 7 = 45 days

Therefore, the expected duration is:

45 days

Explanation:

The Critical Path was determined by comparing the durations of all paths through the network and identifying the longest path. T1 → T3 → T4 → T5 → T6 → T7 has a total duration of 45 days, making it the longest path. All activities on this path have zero slack, meaning any delay in one of these tasks will delay the entire project. T2 has 7 days of slack, so it is not part of the Critical Path.